

Ankush Mondal

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EDUCATION

Techno India University

Bachelor of Technology in Computer Science and Business System

Kolkata, India

Sep. 2022 – Aug 2026

EXPERIENCE

Open Source Contributions

Tursodb, Postgres, FreeBSDOS

- fixed POSIX block-size handling in FreeBSD **df -k** flag. [Link](#) (FreeBSD OS)
- Added **.read** command to execute *SQL* from a file in **TursoDB**. [Link](#) (Tursodb)
- Migrated Turso core from custom SmallVec to **smallvec** crate. [Link](#) (Tursodb)
- Implemented **PostgreSQL protocol logging** to debug dump failures. [Link](#) (Postgres pgagroal)
- update bash script to support **gcc compilation** without coverage. [Link](#) (Postgres pgagroal)
- Fixed **unix domain socket** sharing issue causing high CPU consumption. [Link](#) (Postgres pgagroal)
- Handle ErrorResponse during **SCRAM authentication**. [Link](#) (Postgres pgagroal)
- validate **handle response** status and length edge cases. [Link](#) (Apache Iggy)

Linux System Programming Trainee

Sasken Technologies Ltd (Remote)

June 2025 - Aug 2025

- Developed a monitoring tool in **C** for **Linux systems** to detect kernel memory leaks by parsing the `/proc/slabinfo` and `/proc/vmstat` virtual files.

PROJECTS

Zail: Realtime Filesystem & Journal Monitor | Zig, Linux Syscalls, Epoll, Inotify, [Link](#)

- Built a **low-latency event-driven monitor** using Linux **epoll**, **inotify**, and **signalfd** to track filesystem changes, system signals, and systemd journal updates in a unified loop.
- Engineered a **recursive directory-watch system** with dynamic watch-descriptor mapping and path caching for robust large-tree monitoring.

Mini KVM Hypervisor | C, KVM API, x86 Virtualization

- Built a minimal virtual machine monitor using the **Linux KVM API** to create a VM, map guest physical memory, and execute raw x86 machine code on a virtual CPU.
- Handled **VM exit conditions** (e.g. `KVM_EXIT_HLT`) and inspected vCPU register state, demonstrating low-level control over hardware virtualization.
- Gained hands-on experience with **memory slot registration**, `KVM_RUN` execution loops, and the core KVM `ioctl` interface.

Vec.hpp | C++, Makefile, Memory Management, [Link](#)

- Built a custom `Vec<T>` class with a **chunked paging system** using low-level `mmap`, enabling dynamic expansion without costly reallocations.
- Designed scalable memory architecture with **O(1) element access** via bitwise operations, placement `new`, and pointer arithmetic, minimizing fragmentation.
- Benchmarked up to **500M elements**, achieving a **1.73× speedup** in insertion throughput over `std::vector`.

TECHNICAL SKILLS

Languages & tools: C/C++, Py, go, Rust, SQL, Linux, x86asm, GDB, Makefiles, Bash

Courses: Operating Systems, DBMS, Computer Networks, Theory of Computation

Tech Stack: C, Rust, C++, Linux Kernel, PostgreSQL Internals, Distributed Systems, Low Level Security

Skills: Systems Programming, Memory Management, Data Structures and algorithms, Performance Optimization